

UNIT IV
ASSIGNMENT

Class #8. Symmetric matrices and Diagonalization of the matrix.

1) Orthogonally diagonalize following matrices

$$(i) A = \begin{bmatrix} 16 & -4 \\ -4 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

$$(iii) A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

$$\text{Ans: (i)} P = \begin{bmatrix} \frac{4}{\sqrt{17}} & \frac{1}{\sqrt{17}} \\ \frac{1}{\sqrt{17}} & \frac{4}{\sqrt{17}} \end{bmatrix} D = \begin{bmatrix} 17 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(ii) P = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix} D \rightarrow \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$(iii) P = \begin{bmatrix} \frac{-1}{\sqrt{5}} & \frac{4}{\sqrt{45}} & \frac{-2}{3} \\ \frac{2}{\sqrt{5}} & \frac{2}{\sqrt{45}} & \frac{-1}{3} \\ 0 & \frac{5}{\sqrt{45}} & \frac{2}{3} \end{bmatrix} D \rightarrow \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$